MACHINE LEARNING MINI PROJECT

Detecting Heart Abnormalities Using ECG with CART

CART, SVM, Naive Bayes and Random Forest

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01 · MOTIVATION

Why classify ECG records?

Objective: classify UCI arrhythmia records and compare tree-based and baseline classifiers.

Manual analysis is time-consuming at scale, and rare rhythm classes are difficult to identify.

DATASET

UCI Arrhythmia: 452 patient records · 279 features plus class label · 16 classes, with 13 present locally.

There are 408 missing cells; feature 14 is 83.2% missing. Class 1 has 245 records, while classes 7 and 8 have only 3 and 2.

02 · PIPELINE

From raw ECG data to four model comparisons

One stratified split keeps the evaluation consistent.

01

Load and clean

Load arrhythmia.data (452×280) and convert "?" to missing values.

02

Remove feature 14

Drop feature 14 because 83.2% of its values are missing; 278 features remain.

03

Impute and filter

Median-impute missing values and remove constant features.

04

Split 70 / 30

Stratified seed-42 split: 316 training records and 136 test records.

05

Train and compare

CART, SVM, Naive Bayes, and Random Forest; scale features for SVM.

REFERENCE STUDY

Zhou, Garcia & Rajbhandary
Stanford CS229: CART via R rpart, cp ≈ 0.027, about 80% accuracy. Not a like-for-like comparison.

OUR SETUP

CART + Random Forest
CART ccp_alpha 0.02 by 5-fold CV; Random Forest 100 trees, balanced_subsample, seed 42.

03 · BASELINE

CART reached 63.24% accuracy

10-FOLD CV: 66.59% ± 8.67.

CART reached 63.24% test accuracy

EVIDENCE PANEL · SCREENSHOT PLACEHOLDER

Actual CART notebook output and confusion matrix

Reserved for the final notebook capture. No screenshot was included in the supplied source.

MODEL	CONFIGURATION
CART decision tree	ccp_alpha 0.02 · 5-fold CV selection
SPLIT	10-FOLD CV
316 training · 136 testing	66.59% ± 8.67

Weighted F1: 0.589 · **Macro F1:** 0.237

THE PAPER REPORTS ABOUT 80%, BUT ITS DATA PREPARATION, SPLIT, PRUNING, AND IMPLEMENTATION ARE NOT LIKE-FOR-LIKE WITH THIS COURSEWORK RESULT.

Random Forest reached 66.18% accuracy

EVIDENCE PANEL · SCREENSHOT PLACEHOLDER

Actual Random Forest notebook output and confusion matrix

Reserved for the final notebook capture. No screenshot was included in the supplied source.

66.18%

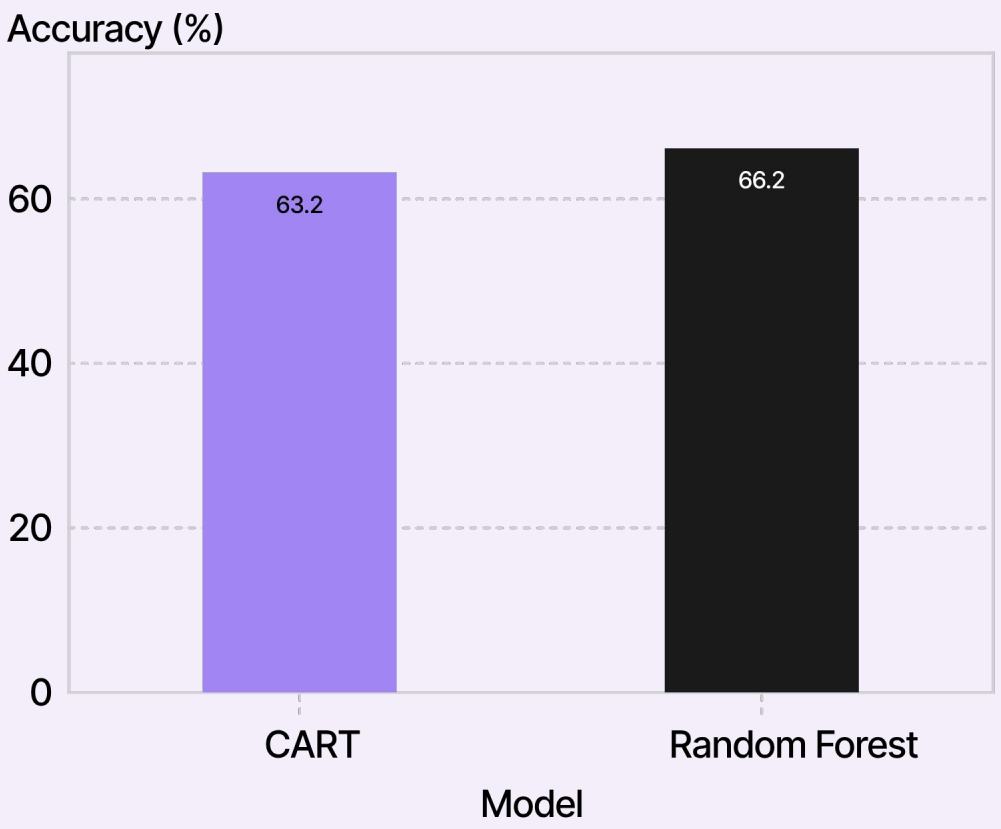
TEST ACCURACY

Ensemble model: 100 trees with balanced_subsample and seed 42.

Same split: 90 of 136 test records correct versus 86 for CART, a +2.9-point accuracy gain.

F1: weighted 0.568 · macro 0.316. Macro F1 improved, while weighted F1 was slightly lower.

Random Forest improved accuracy, not every metric



TEST ACCURACY · 136-RECORD TEST SET

Metric	CART	Random Forest
Accuracy	63.24%	66.18%
Weighted F1	0.589	0.568
Macro F1	0.237	0.316

OTHER BASELINES: SVM WEIGHTED/MACRO F1 0.548/0.252; NAIVE BAYES 0.136/0.130.

Demonstration: the project running end to end

1. Load arrhythmia.data (452×280).
2. Convert “?” values, drop feature 14, impute medians, and remove constants.
3. Apply the stratified 70/30 split with seed 42.
4. Train CART, SVM, Naive Bayes, and Random Forest.
5. Show accuracy, weighted F1, macro F1, and the comparison chart.

Show the actual code execution, preprocessing output, model results, and confusion matrices. Keep the demonstration video within 3 minutes.

Reserve a screenshot strip or gallery for actual dataset, preprocessing, execution, and evaluation captures when provided.

SCREENSHOT PLACEHOLDER 01

Actual dataset and preprocessing output

SCREENSHOT PLACEHOLDER 02

Actual code execution and model results

SCREENSHOT PLACEHOLDER 03

Actual evaluation, confusion matrices, and comparison visuals

Conclusion and next experiments

A modest gain, a careful interpretation.

RANDOM FOREST IMPROVED TEST ACCURACY, BUT RARE CLASSES REMAIN DIFFICULT.

Implemented

Built a reproducible UCI Arrhythmia pipeline with median imputation, constant-feature removal, and a stratified seed-42 split.

Measured

CART: 63.24% accuracy. Random Forest: 66.18%, +2.9 points; 90 versus 86 correct out of 136.

Interpretation

RF macro F1 is higher (0.316 vs 0.237), but weighted F1 is slightly lower (0.568 vs 0.589). Rare classes remain hard.

NEXT EXPERIMENTS

Tune class-aware objectives · inspect rare-class errors · compare additional validation methods · test on additional datasets · investigate feature selection.

LIMITATION

The paper's ~80% result is not like-for-like. This is coursework, not a clinically validated medical diagnostic tool.

Thank You